

Functional Requirements Document (FRD)

Project

Online Travel Reservation Platform (Flights & Hotels)

Version

1.0

Purpose

This document defines the functional requirements for a UK Civil Aviation Authority (CAA) and ATOL compliant travel reservation platform that enables customers to search, compare, book, modify, and cancel flights and hotel reservations.

1. System Modules

Module ID	Module Name
MOD-01	User Management
MOD-02	Flight Search
MOD-03	Hotel Search
MOD-04	Booking Engine
MOD-05	Payment Processing
MOD-06	Customer Account
MOD-07	Notifications
MOD-08	ATOL Compliance
MOD-09	Administration
MOD-10	Reporting & Audit

2. User Registration & Authentication

FR-001 User Registration

Description

Allow customers to create an account.

Inputs

- First Name
- Last Name
- Email Address
- Password
- Mobile Number

Validation

- Email must be unique.
- Password minimum 12 characters.
- Password must contain uppercase, lowercase, number and special character.

Outputs

- User account created.
 - Verification email sent.
-

FR-002 User Login

Description

Allow registered users to access their accounts.

Validation

- Valid credentials required.
- Lock account after 5 failed attempts.

Success Outcome

User redirected to dashboard.

3. Flight Search

FR-003 Flight Search

Description

User searches available flights.

Search Criteria

- Origin Airport
- Destination Airport
- Departure Date
- Return Date
- Passenger Count
- Cabin Class

System Behavior

System calls airline/GDS APIs.

Output

Display matching flights.

FR-004 Flight Results Display

Display:

- Airline
- Flight Number
- Departure Time
- Arrival Time
- Stops
- Duration
- Fare Rules
- Baggage Information
- Total Price

Sorting

- Lowest Price
- Fastest Flight
- Earliest Departure
- Latest Departure

Filters

- Airline
 - Number of Stops
 - Departure Time
 - Arrival Time
 - Price Range
-

FR-005 Flight Selection

User selects flight itinerary.

System shall:

- Revalidate price
- Revalidate seat inventory
- Reserve inventory temporarily

Timeout:

15 minutes

4. Hotel Search

FR-006 Hotel Search

Inputs

- Destination
- Check-in Date
- Check-out Date
- Guests
- Rooms

Output

Available hotels.

FR-007 Hotel Results

Display:

- Hotel Name
- Star Rating
- Room Type
- Amenities
- Images
- Cancellation Policy
- Total Price

Filters

- Price
 - Star Rating
 - Amenities
 - Distance
 - Guest Rating
-

FR-008 Hotel Selection

System shall:

- Verify room availability
- Verify latest pricing
- Lock room inventory

Timeout:

15 minutes

5. Dynamic Package Booking

FR-009 Package Creation

Customer may combine:

- Flight + Hotel

System shall determine:

- Package Holiday

- Linked Travel Arrangement
- Separate Reservations

System shall apply applicable compliance rules.

6. Traveller Information

FR-010 Passenger Details

Collect:

- Title
- First Name
- Last Name
- Date of Birth
- Gender
- Nationality
- Passport Information

Validation

Name must exactly match travel document.

7. Booking Review Screen

FR-011 Booking Summary

Display:

- Flights
- Hotels
- Passenger Details
- Total Cost
- Taxes
- Service Fees
- Cancellation Rules

User must accept:

- Terms & Conditions
 - Privacy Policy
 - Supplier Terms
-

8. Payment Processing

FR-012 Payment Screen

Supported Methods:

- Visa
- Mastercard
- American Express
- Apple Pay
- Google Pay

Validation

3D Secure authentication required.

FR-013 Payment Authorization

System shall:

1. Send payment request.
2. Receive authorization response.
3. Confirm booking only after payment success.

Failure:

Booking not confirmed.

9. Booking Confirmation

FR-014 Booking Creation

Generate:

- Booking Reference
- Supplier Confirmation Number
- Invoice

Store booking record.

10. ATOL Compliance Module

FR-015 Protection Determination

System shall determine whether booking is:

- ATOL Protected
- Not ATOL Protected
- Partially Protected

before payment.

FR-016 ATOL Certificate Generation

Applicable bookings only.

System shall:

- Generate ATOL Certificate
 - Assign Certificate Number
 - Store certificate
 - Email certificate
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FR-017 Financial Protection Notice

Display required protection wording:

- Search Results
 - Booking Summary
 - Confirmation Page
 - Confirmation Email
-

11. Booking Management

FR-018 View Bookings

Users can:

- View itinerary
- View invoices

- Download documents
 - Download ATOL certificates
-

FR-019 Cancellation

System shall:

- Calculate penalties
 - Display refund estimate
 - Obtain customer confirmation
 - Process cancellation
-

FR-020 Refund Processing

System shall:

- Create refund request
 - Track refund status
 - Notify customer
-

12. Notifications

FR-021 Email Notifications

Trigger events:

- Registration
 - Booking Confirmation
 - Cancellation
 - Refund
 - Travel Reminder
-

FR-022 SMS Notifications

Optional:

- Booking Confirmation
 - Flight Reminder
-

13. Administration

FR-023 Admin Dashboard

Admin users can:

- View bookings
 - Search customers
 - Manage content
 - Manage promotions
 - Generate reports
-

FR-024 Customer Service

Support agents can:

- View bookings
 - Add notes
 - Reissue documents
 - Process amendments
-

14. Audit & Compliance

FR-025 Audit Logging

System shall record:

- User
- Timestamp
- Action
- Before Value
- After Value

Logs must be immutable.

FR-026 Regulatory Reporting

Generate:

- ATOL Reports

- Booking Reports
 - Refund Reports
 - Supplier Settlement Reports
-

15. Security Requirements

FR-027 Authentication

- MFA for administrators.
 - Session timeout after 30 minutes.
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FR-028 Data Protection

Encrypt:

- Personal Data
- Passport Data
- Payment Tokens

No card details stored.

16. Error Handling

FR-029 Flight No Longer Available

Display:

"The selected fare is no longer available. Please choose another option."

FR-030 Hotel No Longer Available

Display:

"The selected room is no longer available."

FR-031 Payment Failure

Display failure reason where available.

Allow retry.

17. Integration Requirements

INT-001 Flight Providers

Integrate with:

- Amadeus
 - Sabre
 - Travelport
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INT-002 Hotel Providers

Integrate with:

- Expedia Partner Solutions
 - Hotelbeds
 - Booking Supplier APIs
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INT-003 Payment Providers

Integrate with:

- Stripe
 - Worldpay
 - Adyen
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18. Acceptance Criteria

1. User can search and book flights.
2. User can search and book hotels.
3. User can purchase dynamic packages.
4. Payment is securely processed.
5. ATOL Certificate is generated where required.
6. Customer receives booking documentation.
7. Audit reports are available.
8. Regulatory requirements are satisfied.